

EEG Source Imaging with Brainstorm

Student tutorial · guided source reconstruction and reproducibility exercise

EEG practical exercise using the Wakeman & Henson face-processing dataset

Scope. The introductory lecture discusses both EEG and MEG source imaging. In this practical session, all reconstructions are performed using EEG only. MEG is mentioned for context and for comparison with the broader M/EEG framework.

Learning goals

- Load a Brainstorm protocol and identify the data, forward model, source space and noise model needed for source reconstruction.
- Compute cortical EEG source estimates with minimum-norm imaging.
- Compare three inverse-map variants: current-density MNE, dSPM and sLORETA.
- Explore the effect of regularization by changing the assumed SNR while keeping all other choices fixed.
- Compare source maps around the EEG N170 response and summarize source activity in predefined visual ROIs.
- Compare the conclusions obtained in Brainstorm with those obtained independently by the MNE-Python group.

Dataset and teaching material

Dataset. Wakeman & Henson multimodal face-processing dataset; the hands-on uses subject **sub-01**, Run 01, and the conditions **Famous, Unfamiliar, and Scrambled**.

Dataset paper: <https://www.nature.com/articles/sdata20151>

Frontiers Research Topic: From raw MEG/EEG to publication:

<https://www.frontiersin.org/research-topics/5158/from-raw-megeeg-to-publication-how-to-perform-megeeg-group-analysis-with-free-academic-software/magazine>

PracticalMEEG 2025 Brainstorm materials: <https://zenodo.org/records/17644377>

Prepared single-subject protocol:

https://zenodo.org/records/17644377/files/PracticalMEEG_Single_Precomputed.zip?download=1

1. Install Brainstorm

Brainstorm can be run from MATLAB or as a compiled standalone application. Please install and test it before the hands-on.

Official installation instructions: <https://neuroimage.usc.edu/brainstorm/Installation>

- **MATLAB version:** download the Brainstorm source archive, unzip it, start MATLAB, go to the brainstorm3 folder, and type brainstorm in the command window. Do not add the brainstorm3 folder manually to the MATLAB path.
- **Standalone version:** download the compiled Brainstorm archive and the MATLAB Runtime version specified by the Brainstorm installation page.

- **Database folder:** create a separate `brainstorm_db` folder. Do not place it inside the `brainstorm3` program folder.

2. Load the prepared protocol

1. Download [PracticalMEEG Single Precomputed.zip](#) from the link above. Keep the ZIP file intact.
2. Start Brainstorm.
3. Choose File > Load protocol > Load from zip file.
4. Select `PracticalMEEG_Single_Precomputed.zip`.
5. Wait until the protocol appears in the Brainstorm database explorer.

If Brainstorm reports that the protocol name already exists: rename the ZIP file before importing it, or detach/delete the previous copy from the Brainstorm database before reloading.

3. Orient yourself in the protocol

In the database tree, locate subject **sub-01** and the folder corresponding to **Run 01**. Identify the following ingredients before computing anything:

- EEG averages for Famous, Unfamiliar and Scrambled.
- The channel file / EEG sensor geometry.
- The cortical anatomy and source space.
- The precomputed head model (forward model / lead field).
- The EEG noise covariance matrix, estimated from the prestimulus baseline in the original PracticalMEEG workflow.

Checkpoint. Do not continue until you can point to the three condition averages, the head model and the noise covariance in the database tree.

4. Connect the hands-on to the introductory talk

The hands-on start where the introductory lecture ends. The lecture introduces the ill-posed inverse problem, the forward model, the lead-field matrix, minimum-norm estimation, regularization, depth weighting and normalized variants such as dSPM and sLORETA. Here we keep the data and forward model fixed and test how the inverse solution changes when selected analysis choices change.

Core question. How much of a source-reconstruction result belongs to the data, and how much belongs to the choices we make when solving the inverse problem?

5. Read the Brainstorm source-estimation guide

The relevant Brainstorm documentation is Tutorial 22: Source estimation. Focus on the sections **Source estimation options**, **Computing sources for an average**, **Standardization of source maps**, and **Advanced options: Minimum norm**.

Brainstorm Tutorial 22: Source estimation:

<https://neuroimage.usc.edu/brainstorm/Tutorials/SourceEstimation>

6. Define the parameters

For all comparisons, keep the following choices fixed:

Choice	Use in this hands-on
Subject / run	sub-01 / Run 01
Modality	EEG only
Conditions	Famous, Unfamiliar, Scrambled
Source space	Cortical surface from the prepared protocol
Dipole orientation	Constrained: Normal to cortex
Forward model	Use the precomputed head model supplied in the protocol
Noise model	Use the precomputed EEG noise covariance
Depth weighting / covariance options	Leave unchanged at the Brainstorm/precomputed defaults

Only two factors will be varied: (1) the inverse-method and (2) the regularization parameter expressed as assumed SNR.

7. Compute a first reference solution

Start with the Famous condition. Using the average node produces a result attached directly to that condition, which makes the later comparison of multiple inverse settings easier.

1. Right-click the Famous EEG average.
2. Select Compute sources [2018].
3. Method: Minimum norm imaging.
4. Measure: Current density map.
5. Dipole orientations: Constrained: Normal to cortex.
6. Sensors: EEG.
7. Click Show details and verify Signal-to-noise ratio = 3.
8. Leave the remaining advanced settings unchanged.
9. Run the computation.

To display the result: **right-click the new source file > Cortical activations > Display on cortex**. Open the corresponding EEG average as a time reference. Brainstorm synchronizes the source map and sensor time series.

Visualization only. Amplitude threshold, smoothing and colormap settings change what you see, not the underlying source time series. Keep visualization settings consistent when comparing maps within the same scale.

8. Experiment A — change the inverse method

At a fixed **SNR = 3**, compute the following source estimates for each condition:

Brainstorm option	Short name	SNR	Conditions
Current density map	MNE	3	Famous / Unfamiliar / Scrambled
dSPM	dSPM	3	Famous / Unfamiliar / Scrambled
sLORETA	sLORETA	3	Famous / Unfamiliar / Scrambled

Important: do not compare raw numerical amplitudes across MNE, dSPM and sLORETA as if they were the same quantity. Current-density MNE, dSPM and sLORETA use different normalizations/scales. Across methods, focus on localization, spatial pattern, latency, relative condition effects and ROI time-course shape.

9. Experiment B — change regularization

Keep the inverse method fixed to **dSPM** and compute solutions with three assumed SNR values:

Brainstorm SNR	Equivalent MNE-Python λ^2	Interpretation for exercise
1	1.000	stronger regularization
3	0.111... (1/9)	Brainstorm/MNE classic default
5	0.040	weaker regularization

In Brainstorm: **Compute sources [2018] > Show details > Signal-to-noise ratio**. Brainstorm documents SNR = 3 as the classic default. MNE-Python commonly uses $\lambda^2 = 1/\text{SNR}^2$, which provides a convenient common parameterization for the final cross-toolbox discussion.

Repeat SNR = 1, 3 and 5 for Famous, Unfamiliar and Scrambled. Because dSPM/SNR=3 was already computed in Experiment A, the complete exercise contains 15 unique source maps rather than 18.

10. Inspect the EEG N170

The Wakeman & Henson face dataset shows a characteristic face-related EEG response around **170 ms (N170)**. The corresponding MEG component is often called M170; in this hands-on we analyze EEG, so we refer to N170.

1. Open the source map and corresponding EEG average.
2. Navigate through approximately 150–190 ms and then place the cursor near 170 ms.
3. For a reproducible visual check, use the same smoothing and amplitude-threshold settings within each comparison. A practical starting display is Smooth $\approx$ 30% and Amplitude threshold $\approx$ 20%, as used in the PracticalMEEG walkthrough; remember that these are display parameters only.
4. Record the main cortical region(s) activated around the N170 and whether the map is focal or spatially extended.
5. Compare Famous, Unfamiliar and Scrambled within the same method/SNR before comparing across methods.

Dataset expectation. The PracticalMEEG walkthrough reports a stronger negative EEG deflection for Famous than Scrambled around 170 ms, with later Famous–Unfamiliar differences after about 250 ms. Treat this as a landmark to inspect, not as a result to force.

11. Compare predefined visual ROIs

For the Brainstorm–MNE comparison, use **atlas-defined ROIs** rather than drawing data-driven scouts independently in each toolbox. This keeps the anatomical definition as similar as possible across the two groups.

Suggested bilateral Desikan–Killiany parcels:

ROI	Role in this exercise
Pericalcarine	Early visual cortex / V1 proxy
Lateral occipital	Higher-order occipital visual response; OFA-related region
Fusiform	Ventral visual/face-related response; FFA-related region

1. Display one source file on the cortex.
2. Open the Scout tab.
3. Use the atlas drop-down to select the Desikan–Killiany atlas (or the matching FreeSurfer atlas available in the protocol).
4. Select the left and right parcels of interest.
5. For a polarity-insensitive cross-toolbox summary, use an RMS/power-like ROI summary consistently in both Brainstorm and MNE-Python. If you instead use signed Mean/PCA time series, document the sign-handling choice because cortical orientation can affect the sign.
6. Display the scout time series and inspect 0–300 ms. To compare several conditions, open the corresponding source files and use the Scout display to overlay files.

Brainstorm Tutorial 23: Scouts: <https://neuroimage.usc.edu/brainstorm/Tutorials/Scouts>

12. Record the results

Complete one row for each solution. The aim is to record what is stable and what changes, not to nominate a universal 'best' inverse method.

Method	SNR	Condition	N170 peak/latency	Peak region	Spatial spread	ROI observation

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13. Final Brainstorm × MNE-Python comparison

Bring the two groups back together and compare conclusions in this order:

- **Temporal:** Is the main face-related response located at a similar latency?
- **Spatial:** Are the strongest cortical regions around 170 ms similar?
- **ROI dynamics:** Do the predefined visual ROIs show comparable temporal profiles?
- **Experimental effect:** Is the Famous / Unfamiliar / Scrambled pattern preserved?
- **Method sensitivity:** How much changes when the inverse method changes?
- **Regularization sensitivity:** How much changes when SNR/ λ^2 changes?
- **Software sensitivity:** After matching the main assumptions, are remaining differences between Brainstorm and MNE smaller or larger than the differences induced by analysis choices?

Take-home message. Reproducibility in source imaging is not only about sharing data and code. It requires making the assumptions and parameter choices that shape the inverse solution explicit, inspectable and reproducible.

References and links

[Brainstorm Installation](#)

[Brainstorm Tutorial 22: Source estimation](#)

[Brainstorm Tutorial 23: Scouts](#)

[PracticalMEEG 2025 Brainstorm materials \(Zenodo\)](#)

[PracticalMEEG 2025 Brainstorm walkthrough](#)

[Wakeman & Henson 2015 dataset paper](#)

[MNE-Python source-space SNR example](#)